Software Requirements Specification

for

CESS DocuFlow: A Web-Based Management and Tracking System for the College of Education and Social Sciences, MSU at Naawan

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

<Identify the product whose software requirements are specified in this document, including the revision or release number. Describe the scope of the product that is covered by this SRS, particularly if this SRS describes only part of the system or a single subsystem.>

This Software Requirements Specification (SRS) document defines the complete software requirements for CESS DocuFlow: A Web-Based Document Management and Tracking System for the College of Education and Social Sciences, Mindanao State University at Naawan (abbreviated as CESS DocuFlow), Version 1.0.

The product is a standalone, web-based information system developed as a course project by students of the Bachelor of Science in Information Technology at Mindanao State University at Naawan (MSUN). It is designed to improve and support the existing document processing workflow of the College of Education and Social Sciences (CESS) by replacing the current reliance on physical documents, manual logbooks, Excel sheets, and Google Drive with a centralized digital platform.

The scope of this SRS encompasses the entire system, including:

- All functional modules for the College Secretary (document recording, completeness checking, routing, external office transfer recording, release, and overall monitoring).
- All functional modules for students, including online submission of student documents, file upload, status checking, and receipt of email alerts.
- All functional modules for faculty (CESS and IDS faculty), including online submission of faculty documents, file upload, status checking, receipt of email alerts, and alerts for Notices and Minutes of Meetings.
- All functional modules for reviewing offices (Department Chairpersons, IDS Principal, College Dean, OVCAA, and Office of the Chancellor), including viewing assigned documents and recording endorsement, approvals, or returns with reasons.
- Core supporting features such as user authentication, role-based access control, origin-based routing for CESS and IDS documents, first-come, first-served ordering by date, notifications, reminders, and a centralized processing history.
- Non-functional aspects including performance, security, usability, and maintainability requirements.

This SRS does not cover any external university-wide systems (for example, systems used by the Human Resource Management Office, the Registrar, Budget, or Accounting). Direct

integration with those systems is outside the scope of Version 1.0. The system also does not replace the official approval authority or physical signatures of the designated offices; it records actions and tracks status only.

The system described herein represents the complete deliverable for Version 1.0, intended for use by the College of Education and Social Sciences and the Integrated Development School (MSUN-IDS) connected to CESS.

1.2 Document Conventions

<Describe any standards or typographical conventions that were followed when writing this SRS, such as fonts or highlighting that have special significance. For example, state whether priorities for higher-level requirements are assumed to be inherited by detailed requirements, or whether every requirement statement is to have its own priority.>

This Software Requirements Specification (SRS) follows consistent typographical and structural conventions to ensure clarity readability and ease of maintenance The document is formatted using the following standards:

- Font and Size:
    - Body Text: Book Antiqua, 12-point
    - Headings: Level 1 (chapter titles such as "Introduction"): Bold, 18-point. Level 2 (Section titles such as "Purpose"): Bold, 14-point. Level 3 (subsection titles such as "Description and Priority"):Bold, 11-point.
- Spacing and Alignment:
    - Line spacing: 1.5 for body text; additional spacing before and after headings.
    - Paragraph alignment: Justified for body text; centered for main title and author information.
    - Section and subsection numbers follow a hierarchical decimal format (e.g. 3.1.1).
- Special Formatting:
    - Bold: Used for emphasis, section headings, and key terms on first use.
    - Italic: Used for subsection headings within system features and for placeholders (e.g., TBD).
    - Requirements: Individually numbered with a unique identifier in the format REQ-XX-n (e.g., REQ-OS-01), presented in plain text followed by a colon.
    - System feature names: Written in bold headline style.
- List and Bullets: Used for itemizing functional requirements, stimulus/response sequences, and characteristics.
- Tables and Figure: Referenced in text and captioned appropriately.

- Priority Handling: Each system feature is explicitly assigned a priority level (High, Medium, or Low). All detailed functional requirements under a feature inherit the priority of their parent feature unless otherwise specified.
- Glossary Terms: Key terms are defined in Appendix A. On first significant use in the document, they are presented in plain text without special highlighting.

These conventions align with common academic and IEEE-inspired SRS templates while maintaining consistency with the formatting observed in the associated project proposal document.

1.3 Intended Audience and Reading Suggestions

<Describe the different types of reader that the document is intended for, such as developers, project managers, marketing staff, users, testers, and documentation writers. Describe what the rest of this SRS contains and how it is organized. Suggest a sequence for reading the document, beginning with the overview sections and proceeding through the sections that are most pertinent to each reader type.>

This SRS is primarily intended for the following types of readers:

- Developers (frontend, backend, and database): to know what to build.
- Project Manager, Product Owner, and Team Leader: to plan the work, set priorities, and check scope.
- UI/UX Designers: To design the interface for each user class.
- QA/Tester: to write test cases and verify that each requirement is met.
- Devops: To prepare deployment, environments, and backups.
- Faculty Adviser: members of the Department of Information Technology at Mindanao State University at Naawan who evaluate the project for academic requirements.
- End-Users and Stakeholders:
    - The College Secretary, who served as the primary document controller.
    - Student, CESS faculty, and IDS Faculty who submit document requests.
    - Department Chairpersons, the IDS principal, the College Dean, and other reviewing offices.

The document does not target marketing staff or extend commercial audiences, as this is an internal academic/university system.

Document Organization

This SRS is organized according to standard IEEE-inspired structure:

- Introduction (Section 1.1-1.5): Provides purpose, scope, conventions, audience, and references.
- Overall Description (Section 2): Describes product perspective, features, user classes, operating environment, constraints, assumptions, and high-level scope.
- System Features (Section 3): Details the major functional features with priorities, responses, and traceable requirements.
- External Interface Requirements (Section 4): Covers user interfaces, hardware interfaces, software interfaces, and communication interfaces.
- Other Nonfunctional Requirements (Section 5): Addresses performance, security, quality attributes, and other requirements.
- Appendices: Include glossary, analysis models (diagram), and open issues.

Recommended Reading Sequences

Readers are encouraged to begin with the overview sections for context, then proceed to sections most relevant to their role:

- All readers (including evaluators, stakeholders, and end-users): Start with Introduction (1) and Overall Description (2) to understand the project's purpose, scope, benefits, and high-level features. Review Project Scope (1.4) and Product Features (2.2) for a quick overview.
- End-users (College Secretary, requesters, and reviewing offices): After the overview, focus on User Classes and Characteristics (2.3), User Interfaces (4.1), and System Features (3) to understand available functionality and work flows. Refer to Appendix A: Glossary for terminology.
- Developers and Testers: Read the entire document sequentially. Pay special attention to System Features (3) for detailed functional requirements, External Interface Requirements (4) for technical interfaces, and Other Nonfunctional Requirements (5) for quality and performance criteria. Use Appendix B: Analysis Models as primary reference during implementation and testing.
- Project Evaluators: Begin with introduction and Overall Description, then examine System Features (3) and Appendices to assess requirement completeness, traceability, and alignment with the project proposal.

This structural approach allows readers to gain a comprehensive understanding efficiently while focusing on sections most pertinent to their needs.

1.4 Project Scope

<Provide a short description of the software being specified and its purpose, including relevant benefits, objectives, and goals. Relate the software to corporate goals or business strategies. If a separate vision and scope document is available, refer to it rather than duplicating its contents here. An SRS that specifies the next release of an evolving product should contain its own scope statement as a subset of the long-term strategic product vision.>

The software specified in this SRS is CESS DocuFlow, a web-based centralized platform designed to improve the recording, monitoring, and tracking of documents processed by the College of Education and Social Sciences (CESS) of Mindanao State University at Naawan.

The primary purpose of CESS DocuFlow is to address the current challenges at physical document handling, manual recordkeeping in logbooks, Excel sheets, and Google Drive, lack of real-time status visibility, difficulty in locating documents (especially IDS documents with extra hand-offs), incomplete submissions, and the absence of a clear order for handling requests and of reminders. Students and faculty currently need to visit the College office repeatedly to inquire about the status of their requests. The system provides a single place where every document is recorded once, its movement between offices is visible, and requesters can check progress and receive email alerts.

The system covers the following major functions: online submission of document requests with required forms, attachments and templates; completeness checking by the College Secretary; document recording and centralized repository; status tracking and workflow monitoring; origin-based routing that applies the correct path for CESS documents and for IDS documents (including the IDS Principal reviews); recording of transfers to and from external offices (HRMO, Registrar, Budget, or Accounting) by the College Secretary; first -come, first-served ordering of requests by date and email notifications including reminders and updates every 3 days and alerts to faculty for Notices and Minutes of Meetings; and role-based access for requesters, the College Secretary, the IDS Principal, Department Chairpersons, The College Dean, and other reviewing offices.

The system's effectiveness depends on the accuracy and timeliness of the information entered by authorized users. Final decisions and approvals remain the responsibility of the designated personnel and offices. Direct access and integration with systems used by HRMO, Registrar, Budget, and Accounting are outside the scope of this version. The project is also limited by the development period, available resources, and the requirements established during consultation with the intended users.

The document types, required forms and attachments, steps, and standard processing times used in this SRS come from the client interviews and from the CESS Manual Operations

(2025), which list the services of the college under the Anti-Red Tape (Republic Act No. 9485). These are summarized in Appendix B.

1.5 References

<List any other documents or Web addresses to which this SRS refers. These may include user interface style guides, contracts, standards, system requirements specifications, use case documents, or a vision and scope document. Provide enough information so that the reader could access a copy of each reference, including title, author, version number, date, and source or location.>

2. Overall Description

2.1 Product Perspective

<Describe the context and origin of the product being specified in this SRS. For example, state whether this product is a follow-on member of a product family, a replacement for certain existing systems, or a new, self-contained product. If the SRS defines a component of a larger system, relate the requirements of the larger system to the functionality of this software and identify interfaces between the two. A simple diagram that shows the major components of the overall system, subsystem interconnections, and external interfaces can be helpful.>

CESS DocuFlow is a new, self-contained web application. It is not a follow-on of an existing software product. It replaces the current manual process that uses a logbook, Excel sheet, Google Drive, and a filing cabinet (see Figure 1 of the Project proposal). It is made for a single college, with the College Secretary as the document controller. The system is not part of a larger University and has no direct connection to other MSU at Naawan systems. HRMO, the Registrar, and Budget and Accounting do not use the system. The College Secretary records documents sent to and returned from these offices on their behalf.

The system follows a Web client, API server, and Supabase architecture (see Figure 3 of the Project proposal). The main components are:

Component	**Technology**	**Role**
Web client	React, Typescript, Tailwind CSS (on Vercel)	Interface that users work with in the browser.
API server	Node.js, Express.js (on Vercel)	Handles requests from the Web client over HTTPS (JSON). Applies the business rules. Reads and writes the database, authentication, and file storage. Send email alerts.
Database	PostgreSQL (Supabase)	Storage users, documents, routes,

		status history, and notification flags.
Authentication	Supabase Auth	Login and role-based access.
File Storage	Supabase Storage	Stores uploaded scans and attachments.
Email notification	Email service (TBD)	Sends email alerts and reminders.
Version control and deployment	Git, GitHub, Vercel	Code is kept on GitHub and deployed automatically to Vercel.

The system supports the document services of the college that are listed in the Manual of Operations, such as endorsement letters, faculty documents (teaching load, service record, IPCR, travel order, leave), and student academic documents (INC Form, leave of absence, thesis forms, permit to hold student activities). It records and tracks these documents. It does not change the steps or approvals set by the Manual.

2.2 Product Features

<Summarize the major features the product contains or the significant functions that it performs or lets the user perform. Details will be provided in Section 3, so only a high level summary is needed here. Organize the functions to make them understandable to any reader of the SRS. A picture of the major groups of related requirements and how they relate, such as a top level data flow diagram or a class diagram, is often effective.>

The major features of CESS DocuFlow are summarized below. Detailed functional requirements for each feature are provided in Section 3.

- User Accounts and Roles: Login and role-based access for students, faculty, the IDS Principal, the College Secretary, and the reviewing offices.
- Online Document Submission: Students and Faculties submit requests and upload digital copies.
- Required Forms and Completeness Check: Each document type shows its required forms and attachments, and the College Secretary records a completeness check. A template is available for each document type.
- Document Recording and Central Repository: One record for each document, with Control Number, Revision Number, attachments, and processing history. Records can be searched.
- Origin-Based Routing and Office Actions: The system applies the correct route based on document type and origin (CESS or IDS). Each office endorses, approves, or returns the documents with reasons.

- Document Status Tracking and Processing History: The current status, the office holding the document, and its full movement are visible.
- Requester Progress Monitoring: Requesters check the progress of their own documents online.
- External Office Transfer: The College Secretary records documents sent to and returned from HRMO, the Registrar, and Budget and Accounting.
- Request Order by Date: Requests are handled by the date and time they were submitted, so the first request is handled first.
- Email Notifications and Reminders: Email alerts when a document is returned or ready, reminders and updates every 3 days for pending items, and alerts to faculty for Notices and Minutes of Meetings.
- Workflow Monitoring and Delay Visibility: Dashboards that show pending and delayed.
- Release and Archive: The College Secretary marks documents ready, records the release, and archives the record.
- CESS Homepage and Faculty, Subject, and Syllabus List: A homepage about the College and a list of faculty, the subject they handle, and a syllabus of each subject.

2.3 User Classes and Characteristics

<Identify the various user classes that you anticipate will use this product. User classes may be differentiated based on frequency of use, subset of product functions used, technical expertise, security or privilege levels, educational level, or experience. Describe the pertinent characteristics of each user class. Certain requirements may pertain only to certain user classes. Distinguish the favored user classes from those who are less important to satisfy.>

The system serves several distinct user classes. These classes are differentiated by their roles in the document process, the frequency they use, the subset of functions they need, and their privilege levels.

User Class	**Classification**	**Use/Expertise**	**Main functions used**
College Secretary	Primary User and System Admin	Daily use. Moderate computer skills. Most favored user class.	Completeness check, logging, routing, external transfers, release and archive, dashboards, search, user accounts, document types, and routes, homepage, faculty, subject, and syllabus list, and document templates.

Student	Secondary User (Requesters)	Occasional use. Basic computer skills. Many users.	Submit student document requests, upload files, check status, read return reasons, resubmit.
Faculty (CESS and IDS)	Secondary User (Requesters)	Occasional use. Basic computer skills. Many users.	Submit faculty document requests (IDS faculty follow the IDS route), upload files, check status, read return reasons, resubmit, get alerts for Notices and Minutes.
IDS Principal	Secondary User (Reviewer)	Regular use for IDS documents. Basic skills.	Review and endorse IDS documents (first and second review), submit own documents, check status.
Department Chairpersons, College Dean, OVCAA, Office of the Chancellor.	Secondary User (Reviewing Offices)	Regular to occasional use. Basic skills.	See documents routed to them, review attachments, endorse, approve, or return with reasons, mark whether higher approval is needed.
HRMO, Registrar, Budget and Accounting	External Offices	No access.	None. The College Secretary records their transfers.

2.4 Operating Environment

<Describe the environment in which the software will operate, including the hardware platform, operating system and versions, and any other software components or applications with which it must peacefully coexist.>

CESS DocuFlow is a Web application. It shall operate in the following environment.

User's Devices

- Desktop computers, laptops, tablets and smartphones with an internet connection. No special hardware is needed.
- Suggested minimum device: 4 GB RAM and a screen of 1366 x 768 pixels or higher for computers (TBD).

- Operating systems that can run a current browser: Windows 11 or later, macOS, Linux, Android 8.0 or later, and iOS/IPADOS 13 or later (TBD)
- Browsers: The latest versions of common Web browsers (e.g., Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari). The exact list is TBD.

Server-Side (Cloud)

- Web client: React with TypeScript and Tailwind CSS, hosted on Vercel.
- API server: Node.js with Express.js hosted on Vercel as a serverless deployment.
- Database, Authentication, and File storage: Supabase (PostgreSQL).
- Email: An email service used by the API server to send alerts (provider TBD).
- Development Tools: Git and GitHub, with automatic deployment to Vercel.

The system does not need a specific operating system or special permissions on the user's device, because it runs in the browser. It shall work together with the cloud services listed above and with common office software that users already have (e.g., a PDF viewer to open attachment). Existing Google Drive scans and manual records are not part of the system.

2.5 Design and Implementation Constraints

<Describe any items or issues that will limit the options available to the developers. These might include: corporate or regulatory policies; hardware limitations (timing requirements, memory requirements); interfaces to other applications; specific technologies, tools, and databases to be used; parallel operations; language requirements; communications protocols; security considerations; design conventions or programming standards (for example, if the customer's organization will be responsible for maintaining the delivered software).>

The following items limit the options of the developers:

- Technologies: The team shall use React, Typescript, Tailwind CSS, Node.js, Express.js, Supabase (PostgreSQL), Git, GitHub, and Vercel, as stated in the approved project proposal.
- Development method and time: The team follows an Agile (Kanban) approach with three two-week iterations from October 26 to December 4, 2026. The Design phase is due October 23, 2026 and the Development and Testing and Delivery phases are due December 04, 2026.
- Serveless hosting: The Express API runs on Vercel, so the API shall be stateless and shall use Supabase connection pooling. Deployment shall be tested early.

- Scope limits: the system shall not connect to the systems of HRMO, the Registrar, and Budget and Accounting, and shall not replace the signatures or final decisions of the designated offices.
- Document rules: Documents shall use a Control Number and a Revision Number, as used by the college and shown in the MSUN forms (e.g., Declaration of Action Form).
- Security: Secrets and API keys shall be kept in Vercel environment variables and never in GitHub. Access shall be role-based with Supabase row-level security.
- Code standards: The main branch is protected. Each feature has its own branch, and each change needs a pull request with at least one teammate review before it is merged.
- Resources: The project is limited by the development period, available resources, and the requirements agreed with the intended users.
- Language: The user interface shall be in English.
- Maintenance: The code shall have basic technical documentation so that the College or later student teams can maintain it (TBD).

2.6 User Documentation

<List the user documentation components (such as user manuals, on-line help, and tutorials) that will be delivered along with the software. Identify any known user documentation delivery formats or standards.>

The following user documentation shall be delivered with the software:

- User manual: A step-by-step guide for each user class (students, faculty, College Secretary, and reviewing offices). It is a deliverable of the Testing and Delivery phase.
- Short user guide: A simple guide for staff and students, as stated in the risk mitigation of the Project proposal (adoption).
- Basic technical documentation: A guide for the source code, database, and deployment. It is a deliverable of the Development phase.
- In-system help: Short instructions on screen (e.g., the list of required forms for each document type) and clear error messages.

The manuals shall be delivered as digital files (the format, such as PDF, is TBD). No online tutorial is planned for this release.

2.7 Assumptions and Dependencies

<List any assumed factors (as opposed to known facts) that could affect the requirements stated in the SRS. These could include third-party or commercial components that you plan to use, issues around the development or operating environment, or constraints. The project could

be affected if these assumptions are incorrect, are not shared, or change. Also identify any dependencies the project has on external factors, such as software components that you intend to reuse from another project, unless they are already documented elsewhere (for example, in the vision and scope document or the project plan).>

Assumptions

- The authorized users enter accurate information and update document statuses on time. Documents that are not recorded may not be shown correctly in the system.
- The services, required forms, steps, and standard processing times in the CESS Manual of Operations (2025) are current and are followed by the college.
- The reviewing offices (Department Chairpersons, the IDS Principal, the College Dean, OVCAA, and the Office of the Chancellor) will take part and record their actions in the system.
- Users have internet access and a working university email address that can receive alerts.
- Requesters can provide digital copies (scans or photos) of their forms and attachments.
- Signed paper documents may still be carried between offices. The system records and tracks them, and the College Secretary will release the final document to the requester.
- The College Secretary will act as the System Admin and will set up the document types, requirements, and routes with the client.
- The list of document types and their required forms and attachments will be provided by the College Secretary.

Dependencies

- The system depends on Supabase (database, authentication, and file storage), Vercel (hosting and deployment), GitHub (code), and an email service. If these services are down or change their limits, the system is affected.
- The project depends on the College Dean and the College Secretary for consultation, requirements, and testing.
- The routes of CESS and IDS documents depend on the current college procedures (see Appendix B). A change in the procedures may change the requirements.

3. System Features

<This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product. You may prefer to organize this section by use case, mode of operation, user class, object class, functional hierarchy, or combinations of these, whatever makes the most logical sense for your product.>

No.	System Feature	Code	Priority
3.1	User Accounts and Roles	REQ-AUTH	High
3.2	Online Document Submission	REQ-SUB	High
3.3	Required Form and Completeness Check	REQ-REQT	High
3.4	Document Recording and Central Repository	REQ-REC	High
3.5	Origin-Based Routing and Office Actions	REQ-ROUTE	High
3.6	Document Status Tracking and Processing History	REQ-TRK	High
3.7	Requester Progress Monitoring	REQ-MON	High
3.8	External Office Transfer	REQ-EXT	Medium
3.9	Requests Order by Date	REQ-ORD	Medium
3.10	Email Notifications and Reminders	REQ-NOT	High
3.11	Workflow Monitoring and Delay Visibility	REQ-WFM	Medium
3.12	Release and Archive	REQ-REL	High
3.13	CESS Homepage and Faculty, Subject, and Syllabus List	REQ-INFO	Medium

3.1 User Accounts and Roles

<Don't really say "System Feature 1." State the feature name in just a few words.>

3.1.1 Description and Priority

<Provide a short description of the feature and indicate whether it is of High, Medium, or Low priority. You could also include specific priority component ratings, such as benefit, penalty, cost, and risk (each rated on a relative scale from a low of 1 to a high of 9).>

This feature allow users log in and limits what each person can see and do based on their role. The roles are Student and Faculty (CESS faculty and IDS Faculty; both are requesters), IDS Principal, reviewing Offices (Department Chairperson, College Dean, OVCAA, Office of the Chancellor), and College Secretary (System Admin).

Priority: High

All other features depend on it, and without it the records are not protected.

3.1.2 Stimulus/Response Sequences

<List the sequences of user actions and system responses that stimulate the behavior defined for this feature. These will correspond to the dialog elements associated with use cases.>

- The user opens the system and enters an email and password. The system checks them and opens the home page for the user's role. If they are wrong, the system shows an error message.
- The user clicks Log out, or is inactive for a long time. The system ends the session.
- The system admin creates, edits, or deactivates an account and assigns a role. The system saves it and the user gets access based on the role.
- The user tries to open a page or records that are not allowed for the role. The system denies access and shows a message.

3.1.3 Functional Requirements

<Itemize the detailed functional requirements associated with this feature. These are the software capabilities that must be present in order for the user to carry out the services provided by the feature, or to execute the use case. Include how the product should respond to anticipated error conditions or invalid inputs. Requirements should be concise, complete, unambiguous, verifiable, and necessary. Use "TBD" as a placeholder to indicate when necessary information is not yet available.>

<Each requirement should be uniquely identified with a sequence number or a meaningful tag of some kind.>

REQ-AUTH-1: The system shall require login before any page or record can be opened.

REQ-AUTH-2: The system shall support these roles: Student, Faculty (CESS or IDS), IDS Principal, Department Chairperson, College Dean, OVCAA, Office of the Chancellor, and College Secretary (System Admin). Students and Faculty are separate roles with their own pages and document types.

REQ-AUTH-3: The System Admin shall be able to create, edit, deactivate, and assign roles to user accounts. How student and faculty accounts are registered is TBD.

REQ-AUTH-4: The system shall show each user only the pages, records, and buttons allowed for their role. Requesters see only their own documents, reviewing offices see documents routed to their office, and the College Secretary sees all documents.

REQ-AUTH-5: The system shall not require HRMO, the Registrar, or Budget and Accounting to have accounts.

REQ-AUTH-6: The system shall allow users to log out and shall end a session after a period of no activity (length TBD).

REQ-AUTH-7: The system shall allow users to reset their password through their registered university email.

REQ-AUTH-8: The system shall show a clear message for a failed login or for denied access.

REQ-AUTH-9: The system shall require a university email address for every account. It shall be used to log in and to receive alerts, reminders, and updates.

3.2 Online Document Submission

3.2.1 Description and Priority

This feature allows students, CESS faculty, and IDS Faculty submit document requests online and upload digital copies of the required forms and attachments, without bringing a paper packet to the College office.

Priority: High

It is the starting point of the whole process.

3.2.2 Stimulus/Response Sequences

- The Requester clicks Submit new request and chooses a document type. The system shows the form for that type and the list of required forms and attachments.

- The Requester fills in the details, uploads the files, and clicks the Submit. The system checks the form and files, saves the request, shows a reference number, and sets the status of Submitted.
- The Requester uploads a file that has the wrong type or is too large. The system rejects the file and shows a clear error message.
- The Requester opens a Returned request, reads the reasons, fixes the problems, and resubmits. The system keeps the earlier record and history, increases the Revision number, and sets the status back to Submitted.

3.2.3 Functional Requirements

REQ-SUB-1: The system shall let a requester create a request by choosing a document type from a list maintained by the System Admin. The list shall show only the document types that the role of the user (Student or Faculty) can submit.

REQ-SUB-2: The system shall collect the title or subject, short document type, short description, and the requester's name and office or department (taken from the account). The date and time of submission shall be added automatically.

REQ-SUB-3: The system shall let the requester upload one or more digital copies of the required forms and attachments. Allowed file types are TBD. The proposed maximum size is 10 MB per file.

REQ-SUB-4: The system shall check the required fields, file type, and file size before saving and shall show clear error messages next to the problem.

REQ-SUB-5: The system shall assign a unique Control Number when the request is saved, set the status to Submitted, and show a confirmation to the requester.

REQ-SUB-6: The system shall set the origin of the request (CESS or IDS) from the requester's account.

REQ-SUB-7: The system shall not allow a requester to change a submitted request unless it has been returned.

REQ-SUB-8: The system shall let the requester resubmit a returned request under the same Control Number with a new Revision Number, and shall keep the earlier files and history.

3.3 Required Forms and Completeness Check

3.3.1 Description and Priority

This feature shows the required forms and attachments for each document type, so requesters know what to submit. It also lets the College Secretary check each submission for completeness and record the result. It also provides a template for each document type that requesters can download and fill in.

Priority: High.

It reduces incomplete submissions, which is one of the main problems of the current process.

3.3.2 Stimulus/Response Sequences

- The requester chooses a document type. The system shows the required forms and attachments, with a link to download to template of each document, before the upload.
- The College Secretary opens a new submission. The system shows the uploaded files next to the checklist of requirements.
- The College Secretary marks each item as present or missing and selects Complete. The system records the check (who and when) and allows the document to be logged and routed.
- The College Secretary selects Incomplete and enters the reasons. The system sets the status to Returned and sends an email with the reasons to the requester.

3.3.3 Functional Requirements

REQ-REQT-1: The System Admin shall be able to set and edit the required forms and attachments for each document type. Each document type shall have a checklist.

REQ-REQT-2: The system shall show the required forms and attachments for the chosen document type on the submission form and shall mark each one as required or optional.

REQ-REQT-3: The system shall show the College Secretary a completeness checklist for each new submission.

REQ-REQT-4: The system shall record the result of each completeness check: date and time, who checked, result, and remarks.

REQ-REQT-5: The system shall require reasons when a submission is marked Incomplete.

REQ-REQT-6: The system shall not allow a document to be routed until it is marked Complete.

REQT-7: The system shall show, for each required form or attachment, where to get it (for example, Registrar's Office or HRMO), as listed in the CESS Manual of Operations.

REQ-REQT-8: The System Admin shall set up the first list of document types and checklists from the Manual of Operations (see Appendix B).

REQ-REQT-9: The system shall provide a template for each document type that requesters can download, fill in, and upload.

REQ-REQT-10: The system admin shall be able to upload and replace the templates. The first templates come from the forms of the college (see Appendix B)

3.4 Document Recording and Central Repository

3.4.1 Description and Priority

This feature keeps every document received by the College Secretary in one central repository, with its submission details, attachments, and processing history. It replaces the logbook, Excel sheets, Google Drive records, and the filing cabinet as the main record.

Priority: High.

It is the main objective of the project.

3.4.2 Stimulus/Response Sequences

- The College Secretary marks a submission Complete and selects Log. The system creates the document record with a Control Number and Revision Number.
- An authorized user enters a keyword or filter (Control Number, title, requester, type, status, office, date). The system shows the matching records the user is allowed to see.
- The user opens a record. The system shows the details, attachments, and processing history.
- The College Secretary opens an archived record. The system shows it in read-only form.

3.4.3 Functional Requirements

REQ-REC-1: The system shall keep one record for each document with: Control Number, Revision Number, document type, title, origin (CESS or IDS), requester, date received, current status, current office, and last-updated time.

REQ-REC-2: The system shall use a Control Number and a Revision Number to track and refer to a document. The number format shall follow the college's format (MSU-[office]-COM-[number]-[year]-REV00, as given by the College Secretary on October 16, 2026). The year changes by document, and the revision part increases after each resubmission (e.g., REV01). Whether the number is made by the System or typed by the College Secretary is TBD.

REQ-REC-3: The system shall store uploaded files in file storage and link them to the document record.

REQ-REC-4: The system shall keep a processing history for every document (office, action, reasons or remarks, who, date and time) and shall not delete it.

REQ-REC-5: The system shall let authorized users search and filter records by Control Number, title, requester name, document type, status, office, and date range.

REQ-REC-6: The system shall let authorized users download attachments so that e-copies can be sent for audit.

REQ-REC-7: The system shall not hard-delete document records or their files. The retention period is TBD.

REQ-REC-8: The system shall let the College Secretary create a record for a document received in person, marked as encoded by the Secretary (to be confirmed with the client).

3.5 Origin-Based Routing and Office Actions

3.5.1 Description and Priority

This feature applies the correct approval path based on the document type and its origin (CESS or IDS). Each office records its action as endorsed, approved, or returned with reasons, so every document follows the right sequence of offices.

Priority: High.

It supports the main workflow and removes the need to set the route by hand each time.

3.5.2 Stimulus/Response Sequences

- The College Secretary marks a document Complete and selects Log and route. The system builds the route from the document type and origin, sets the first office as the current office, and shows the document in that office's list.
- A reviewing office opens its list, reviews the document and attachments, and clicks Endorse or Approve. The system saves the action, moves the document to the next office in the route, and updates the status.
- A reviewing office clicks Return and enters the reasons. The system sets the status to Returned, stops the route, and sends an email with the reasons to the requester.
- The College Dean marks that the document needs higher approval. The system continues the route to the OVCAA and the Office of the Chancellor.
- The last office in the route completes its action. The system sets the status to Ready for Release.

3.5.3 Functional Requirements

REQ-ROUTE-1: The system shall store the route (an ordered list of offices) for each combination of document type and origin. The System Admin shall be able to edit routes without changing the code.

REQ-ROUTE-2: For CESS documents the default route shall be: College Secretary, Department Chairperson, College Dean, then, when required, OVCAA and the Office of the Chancellor.

REQ-ROUTE-3: For IDS documents the route shall be: College Secretary, Department Chairperson, IDS Principal, College Dean, then any other required office. The IDS Principal's first and second review shall be supported (exact sequence TBD).

REQ-ROUTE-4: The system shall select the route automatically from the document type and origin. The College Secretary shall not need to set it by hand.

REQ-ROUTE-5: The system shall let the College Dean mark whether a document needs higher approval, and shall route it to OVCAA and the Office of the Chancellor when it does.

REQ-ROUTE-6: The system shall let each office record one of these actions: Endorsed, Approved, or Returned. Reasons shall be required for Return.

REQ-ROUTE-7: The system shall let only the office that currently holds a document record an action on it.

REQ-ROUTE-8: After each action, the system shall update the status, current office, processing history, and last-updated time.

REQ-ROUTE-9: The system shall send a returned document back to the requester. After resubmission, the document shall go back to the College Secretary's completeness check, under the same Control Number with a new Revision Number. (see REQ-SUB-8).

REQ-ROUTE-10: The system shall record the decisions of the offices only. It shall not replace their signatures or final decisions.

REQ-ROUTE-11: The first routes in the system shall be set up from the client interview and the Manual of Operations (see Appendix B), and shall be confirmed with the College Secretary.

3.6 Document Status Tracking and Processing History

3.6.1 Description and Priority

This feature shows the current status of every document, the office that holds it, and the full history of its movement between offices.

Priority: High.

It solves the problem of not knowing where a document is or who last handled it.

3.6.2 Stimulus/Response Sequences

- An office or the College Secretary records an action. The system updates the status, the current office, and the time right away.
- A user opens a document record. The system shows the current status, the current office, the time the document has been waiting there, and a timeline of all steps.
- A user opens a list of documents. The system shows the status and current office of each document.

3.6.3 Functional Requirements

REQ-TRK-1: The system shall use a fixed list of statuses (see Appendix B), including Submitted, Returned, With an office, With an external office, Ready for Release, and Released.

REQ-TRK-2: The system shall show for each document the current status, the current office, the last-updated time, and the time it has waited in the current office.

REQ-TRK-3: The system shall show the processing history in order, with the office, action, reasons or remarks, who recorded it, and the date and time.

REQ-TRK-4: The system shall show users the updated status and location without delay after an action is recorded (see Section 5.1).

3.7 Requester Progress Monitoring

3.7.1 Description and Priority

This feature lets students, CESS faculty, and IDS faculty check the progress of their submitted documents online, so they do not have to visit the college office or follow up by email or messenger.

Priority: High.

It is a main benefit for requesters.

3.7.2 Stimulus/Response Sequences

- The requester logs in and opens My Documents. The system shows the list of the requester's documents with their status.
- The requester opens a document. The system shows the current office, the timeline, and what to do next.
- The document was returned.The system highlights it and shows the reasonsand the missing requirements.
- The document is ready. The system highlights it and shows that it can be collected from the College Secretary.

3.7.3 Functional Requirements

REQ-MON-1: The system shall list all requests of the logged-in requester with Control Number, document type, date submitted, and status.

REQ-MON-2: The system shall highlight Returned and Ready for Release documents.

REQ-MON-3: The system shall show the reasons for a return and the missing requirements in the requester's document view, and not only by email (see REQ-NOT-1).

REQ-MON-4: The system shall let a requester search and filter their documents by status, document type, and date.

3.8 External Office Transfers

3.8.1 Description and Priority

This feature lets the College Secretary record documents sent to and returned from HRMO, the Registrar, and Budget and Accounting. These offices do not have accounts, so the Secretary records the hand-off and returns on their behalf.

Priority: Medium.

It keeps a document visible in the system while it is outside the reviewing offices.

3.8.2 Stimulus/Response Sequences

- The College Secretary opens a document and selects Send to the external office. The Secretary chooses the office and enters the date and remarks. The system sets the location to that office and adds the step to the history.
- The external office returns the document. The College Secretary selects Record return and enters the date and remarks. The system moves the document back to the route and updates the history.
- The College Secretary opens the list of pending external transfers. The system shows each document, the office, and the number of days waiting.

3.8.3 Functional Requirements

REQ-EXT-1: The system shall let the College Secretary record the transfer of a document to HRMO, the Registrar, or Budget and Accounting. The list of external offices shall be editable by the System Admin. Only the College Secretary shall record external transfers.

REQ-EXT-2: The system shall record the office, date sent, who handed over the document, and remarks.

REQ-EXT-3: The system shall let the College Secretary record the return of the document with the date returned and remarks.

REQ-EXT-4: The system shall show the status With external office (name of the office) and the days waiting while the document is outside.

REQ-EXT-5: The system shall let only the College Secretary record external transfers.

REQ-EXT-6: The system shall show a list of documents pending in external offices for follow-up.

REQ-EXT-7: The system shall not require external offices to log in or connect to their own systems.

3.9 Request Order by Date

3.9.1 Description and Priority

This feature handles requests by the date and time they were submitted. The request that came first is handled first. There are no priority flags or levels.

Priority: Medium.

It keeps the order of handling fair and simple, and older requests are not forgotten.

3.9.2 Stimulus/Response Sequences

- A requester submits a request. The system saves the date and time of submission, which sets the place of the request in the order.
- An office or the College Secretary opens a list. The system shows the oldest submitted requests first.
- A user sorts a list by date submitted. The system shows the list from oldest to newest or from newest to oldest.

3.9.3 Functional Requirements

REQ-ORD-1: The system shall order the lists of the College Secretary and the reviewing offices by date and time submitted, with the oldest request first by default. Which date is used for a resubmitted document is TBD.

REQ-ORD-2: The system shall let users sort lists by date submitted, from oldest to newest or from newest to oldest.

REQ-ORD-3: The system shall not use priority levels or flags. The order shall be based only on the date and time submitted.

3.10 Email Notifications and Reminders

3.10.1 Description and Priority

This feature sends email alerts to requesters when their document is returned or ready, sends reminders and updates every 3 days while a document is pending without action. It also tells faculty by email when a Notice or Minutes is sent to them.

Priority: High.

It removes the need for repeated in-person follow-ups.

3.10.2 Stimulus/Response Sequences

- A document is returned. The system sends an email to the requester with the document reference, the office that returned it, and the reasons.
- A document is ready for release. The system sends an email to the requester.
- A document stays in an office without action. Every 3 days, the system sends a reminder to that office, sends an update to the requester, and shows the document as pending on the dashboard.
- A Notice or Minutes is released to the faculty. The system sends an email to each faculty member who should receive it, with a link to open it.
- An email cannot be sent. The system records the failure and lets the College Secretary try again.

3.10.3 Functional Requirements

REQ-NOT-1: The system shall send an email to the requester when a document is returned. The email shall include the Control Number, the office that returned it, and the reasons.

REQ-NOT-2: The system shall send an email to the requester when a document is ready for release.

REQ-NOT-3: The system shall send a reminder to the office that holds a pending document, and an update to the requester, every 3 days until an action is recorded (as asked by the College Secretary on October 16, 2026). The update shall show the office that holds the document and the days waiting. The System Admin shall be able to change the number of days.

REQ-NOT-4: The system shall log every notification (recipient, type, time, and result) and shall allow a retry when sending fails.

REQ-NOT-5: The email text shall be short and plain, and shall include a link to the document in the system.

REQ-NOT-6: The system shall send an email to faculty when a Notice or Minutes of Meeting is released to them, with a link to open it in the system.

3.11 Workflow Monitoring and Delay Visibility

3.11.1 Description and Priority

This feature gives the College Secretary and the reviewing offices a dashboard of pending requests, delayed documents, and documents that need follow-up.

Priority: Medium.

It helps staff find delays early.

3.11.2 Stimulus/Response Sequences

- The College Secretary opens the dashboard. The system shows the number of documents by status and office, and lists of pending, delayed, and external documents.
- A reviewing office opens its dashboard. The system shows the documents waiting for its action, with the days waiting.
- The user selects a document in a list. The system opens its record.
- The user applies a filter. The system updates the lists.

3.11.3 Functional Requirements

REQ-WFM-1: The system shall give the College Secretary a dashboard showing the number of documents by status and by office.

REQ-WFM-2: The system shall show pending documents with the number of days waiting.

REQ-WFM-3: The system shall list documents that need follow-up: delayed and pending in an external office.

REQ-WFM-4: The system shall give each reviewing office a list of the documents waiting for its action.

REQ-WFM-5: The system shall let users filter by document type, origin, office, and date range.

REQ-WFM-6: The system shall show the hand-offs of IDS documents between the IDS Principal, the College Secretary, and the Department Chairpersons in the same view.

REQ-WFM-7: The system shall store a standard processing time (in working days) for each document type, taken from the Manual of Operations (see Appendix B). The System Admin shall be able to change it.

REQ-WFM-8: The system shall mark a document as Delayed when it has waited longer than the upper limit of its standard processing time, and shall show the Delayed documents on the dashboard.

REQ-WFM-9: The system may let the College Secretary export a list of records for a chosen period (format TBD).

3.12 Release and Archive

3.12.1 Description and Priority

This feature lets the College Secretary mark a completed document as ready, release it to the requester, and archive the record.

Priority: High.

It completes the process from submission to release.

3.12.2 Stimulus/Response Sequences

- The route is complete. The system sets the status to Ready for Release and sends an email to the requester.
- The requester collects the document. The College Secretary records the release with the date and the name of the person who received it. The system sets the status to Released.
- The system moves the released document to the archive. Users can still search and open the record, but cannot change it.

3.12.3 Functional Requirements

REQ-REL-1: The system shall let only the College Secretary mark a document Ready for Release and Released.

REQ-REL-3: The system shall record the release date and the name of the person who received the document.

REQ-REL-2: The system shall record the release date and the name of the person who received the document.

REQ-REL-3: The system shall archive released documents. Archived records shall be read-only and searchable.

REQ-REL-4: The system shall let the College Secretary attach a scanned copy of the final signed document to the record.

3.13 CESS Homepage and Faculty, Subject, Syllabus List

3.13.1 Description and Priority

This feature gives the system a homepage about the College of Education and Social Sciences and a list of the faculty, the subjects they handle, and the syllabus of each subject. The College Secretary asked for the homepage and the faculty and subject list during the interview on October 7, 2026, and for the syllabus list on October 16, 2026.

Priority: Medium.

They are client requirements but are not part of the main document workflow.

3.13.2 Stimulus/Response Sequences

- A visitor opens the system address. The system shows the homepage about CESS and a link to log in.
- A logged-in user opens Faculty and Subjects. The system shows the list of faculty and the subjects they handle.
- A logged-in user opens a subject in the list. The system shows the syllabus of that subject and lets the user open or download it.
- The College Secretary adds, edits, or removes an item in the list, including the syllabus of a subject. The system saves the change.

3.13.3 Functional Requirements

REQ-INFO-1: The system shall have a homepage that gives information about CESS and a link to log in. The content comes from the client (TBD).

REQ-INFO-2: The system shall have a list of faculty, the subjects they handle, and the syllabus of each subject.

REQ-INFO-3: The system shall let the College Secretary add, edit, and remove faculty, subjects, and syllabus files in the list. Whether a syllabus is linked to the Syllabus document that goes through the Chairperson and the Dean is TBD.

REQ-INFO-4: The system shall let logged-in users view the list and open or download the syllabus of a subject. Whether the list is also public is TBD.

4. External Interface Requirements

4.1 User Interfaces

<Describe the logical characteristics of each interface between the software product and the users. This may include sample screen images, any GUI standards or product family style guides that are to be followed, screen layout constraints, standard buttons and functions (e.g., help) that will appear on every screen, keyboard shortcuts, error message display standards, and so on. Define the software components for which a user interface is needed. Details of the user interface design should be documented in a separate user interface specification.>

4.2 Hardware Interfaces

<Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the nature of the data and control interactions between the software and the hardware, and communication protocols to be used.>

CESS DocuFlow has no direct hardware interface. It does not control or read any special device. Users reach the system through standard hardware with a web browser and an internet connection: desktop computers, laptops, tablets, and smartphones.

Requesters may use a scanner or a phone camera to create digital copies of their documents, but the system only receives the uploaded files over HTTPS. Users may also use a printer through their browser to print a page, if they need to. The servers are provided by Vercel and Supabase as cloud services, so the team does not manage physical server hardware.

4.3 Software Interfaces

<Describe the connections between this product and other specific software components (name and version), including databases, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Refer to documents that describe detailed application programming interface protocols. Identify data that will be shared across software components. If the data sharing mechanism must be implemented in a specific way (for example, use of a global data area in a multitasking operating system), specify this as an implementation constraint.>

4.4 Communications Interfaces

<Describe the requirements associated with any communications functions required by this product, including e-mail, web browser, network server communications protocols, electronic forms, and so on. Define any pertinent message formatting. Identify any communication standards that will be used, such as FTP or HTTP. Specify any communication security or encryption issues, data transfer rates, and synchronization mechanisms.>

The system uses the following communication functions:

- Web browser: users access the system through a web browser using HTTPS.
- Client and API: the web client and the API server exchange JSON over HTTPS using REST endpoints. All traffic shall be encrypted with HTTPS (TLS).
- API and Supabase: the API server connects to the Supabase database, authentication, and file storage over secure connections, using connection pooling.
- File upload: files are sent from the browser to the system over HTTPS. Allowed file types and the maximum size are checked before saving (TBD, for example PDF and image files).
- Email: the API server sends email alerts and reminders through the email service. Each email has a subject, a short plain message with the Control Number, the status or reason, and a link to the system. Emails shall not contain attachments or more personal data than needed. Emails are sent to the university email address of the user.
- Updates: when a status changes, other users shall see the update in the system within a few seconds after refreshing or loading the page. Real-time updates may be used where Supabase supports them.

Data transfer speeds depend on the user's internet connection. The target page response times are in Section 5.1.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

<If there are performance requirements for the product under various circumstances, state them here and explain their rationale, to help the developers understand the intent and make suitable design choices. Specify the timing relationships for real time systems. Make such requirements as specific as possible. You may need to state performance requirements for individual functional requirements or features.>

The performance targets below are written to help developers make design choices. The numbers are proposed targets and shall be checked during testing and confirmed with the client (see Appendix C).

- **Page load:** common pages (login, My Documents, dashboard, and document details) shall load in 3 seconds or less on a normal campus internet connection.
- **Actions:** saving an action (for example, Endorse, Return, or a status update) shall give a confirmation in 2 seconds or less. Status updates shall take one or two clicks.
- **Search:** search results shall appear in 3 seconds or less for the expected number of records (number TBD).
- **File upload:** a file of up to 10 MB shall upload without errors (proposed limit, TBD), and the user shall see the upload progress.
- **Notifications:** email alerts shall be sent within 5 minutes after the event. Failed emails shall be retried.
- **Users:** the system shall support the expected number of users at the same time without slow response (target of at least 50 users at the same time, TBD).
- **Hosting:** because the API runs as a serverless function on Vercel, the first request after a quiet period may be slower. This shall be tested early.

5.2 Safety Requirements

<Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. Define any safeguards or actions that must be taken, as well as actions that must be prevented. Refer to any external policies or regulations that state safety issues that affect the product's design or use. Define any safety certifications that must be satisfied.>

The system is a software-only product and does not control any physical device, so there is no risk of physical harm. The safety requirements are about loss or damage to records and wrong handling of documents:

- Loss of data: the database shall be backed up regularly with Supabase backups. A restore shall be tested. The status history shall allow records to be traced and recovered.
- Accidental deletion: documents and history shall not be hard-deleted. Records shall be archived instead. Actions such as Return and Release shall ask for confirmation.
- Wrong routing or release: the route shall be set automatically from the document type and origin. Only the office holding a document can act on it, and only the College Secretary can release it.
- Wrong decisions: the system shall show clearly that it only records the decisions of the offices. Signatures and final approvals stay with the designated offices.
- Wrong information: the system shall show the last-updated time on every record, so users can see if the information is old.

The system shall handle personal data according to the Data Privacy Act of 2012 (Republic Act No. 10173). No safety certification is required for this release.

5.3 Security Requirements

<Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements. Refer to any external policies or regulations containing security issues that affect the product. Define any security or privacy certifications that must be satisfied.>

The following security and privacy requirements apply:

- Authentication: every user shall log in with an individual account. Passwords shall be stored securely by Supabase Auth and shall not be visible to anyone.
- Role-based access: access shall depend on the user's role (students, faculty, College Secretary, and reviewing offices). Supabase row-level security shall be used so that users can only read and change the records they are allowed to. Every role's permissions shall be tested before release.
- Protection of files: uploaded files shall be stored in private storage and shall be opened only by authorized users through secure links.
- Encryption: all communication between the browser, the API server, and Supabase shall use HTTPS.

- Secrets: API keys and other secrets shall be kept in Vercel environment variables and never in GitHub. Keys shall be rotated if they are exposed, and code review shall include a secrets check.
- Upload checks: the file type and size shall be validated on upload to avoid invalid or unsafe files.
- Audit trail: the system shall record who did each action and when. The records cannot be changed or removed by users.
- Privacy: the system shall collect only the personal data needed for the process and shall show it only to users who need it (Data Privacy Act of 2012).

5.4 Software Quality Attributes

<Specify any additional quality characteristics for the product that will be important to either the customers or the developers. Some to consider are: adaptability, availability, correctness, flexibility, interoperability, maintainability, portability, reliability, reusability, robustness, testability, and usability. Write these to be specific, quantitative, and verifiable when possible. At the least, clarify the relative preferences for various attributes, such as ease of use over ease of learning.>

The most important quality attributes, in order of importance, are: security and correctness of records first, then usability, reliability, maintainability, and performance. The attributes are written so that they can be checked, and they follow the ISO/IEC 25010 quality criteria that were used to evaluate similar systems.

- Usability: the system shall be easy for the College Secretary to use every day and easy for requesters to learn without training. A requester shall be able to submit a request by following the on-screen steps and the required-forms list. The target is a mean ISO/IEC 25010 rating of at least 4.0 out of 5 from the intended users during testing (TBD).
- Correctness: the status, current office, and history shown shall always match the last action recorded.
- Reliability and availability: the system shall be available during office hours. Backups shall be made regularly. The expected availability (for example, 99%) depends on Vercel and Supabase (TBD).
- Maintainability: the code shall be written in TypeScript and organized in modules. Every change shall have a pull request with a review, and the technical documentation shall be kept up to date.

- Flexibility and adaptability: document types, required forms, routes, external offices, and reminder settings shall be changed by the System Admin without changing the code, so the system can follow changes in the college's procedures.
- Portability: the system shall work on the latest versions of common browsers and on desktop, tablet, and phone screens.
- Testability: each requirement has a tag, so test cases can be traced to it. Developers write unit and API tests, and testers write manual test cases.
- Robustness: the system shall check inputs and show clear error messages without losing the user's data.
- Interoperability: the API uses JSON over REST, so future connection with other systems (for example, the systems of HRMO or the Registrar) is possible later, although it is outside this release.

6. Other Requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

The following other requirements apply to this release:

- Database: the database shall be relational (PostgreSQL on Supabase) and keep data consistent. The main data items are listed in Appendix B. The retention period of records is TBD.
- Existing records: moving old records from the logbook, Excel sheets, and Google Drive into the system is not required for this release (to be confirmed with the client).
- Language and format: the interface is in English. Dates and times use the Philippine time zone (UTC+8).
- Legal: personal data shall be handled according to the Data Privacy Act of 2012.
- Reuse: the system supports and tracks the existing process and uses the college's Control Number and Revision Number. Document types and routes can be changed later for other uses.
- Deployment: code on the main branch shall be deployed automatically to Vercel only after review. Vercel preview builds are used to try changes before they go live.
- Training: a short user guide and user manual shall be given to the College Secretary and the reviewing offices (see Section 2.6).
- Anti-Red Tape Act: the college publishes its services and processing times under Republic Act No. 9485. The system shall help the college follow its standard processing times by showing when a document is late (see REQ-WFM-8).

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Terms	**Meaning**
CESS	College of Education and Social Sciences
MSUN at Naawan/ MSUN	Mindanao State University at Naawan
MSUN-IDS/ IDS	MSU at Naawan - Integrated Development School, a school connected to CESS.
SRS	Software Requirements Specification
OVCAA	Office of the Vice Chancellor for Academic Affairs
OC	Office of the Chancellor
VCAF	Vice Chancellor for Administration and Finance (appears in the endorsement route and the DA form; to be confirmed)
HRMO	Human Resource Management Office
Registrar	The office that keeps student academic records.
Budget and Accounting	The offices that handle budget and funds.
Document controller	The person who receives, records, routes, releases, and files documents. In CESS, this is the College Secretary.
College Secretary	The CESS document controller. In the system, the primary user and System Admin.
System Admin	A user who manages accounts, documents types, required forms, routes, and settings.
Requester	A Student or a Faculty member (CESS or IDS) who submit a document request.
Student	A student who submits student documents (e.g., thesis form)
Faculty	A CESS or IDS faculty member who submits faculty documents and gets alerts for Notices and Minutes.
Reviewing Office	An office that reviews, endorses, approves, or returns documentations in the system (Department Chairperson, College Dean, OVCAA, Office of the Chancellor; the IDS Principal for IDS

	documents).
External Office	An office that is not a system user (HRMO, Registrar, Budget and Accounting)
Origin	Where a document comes from: CESS or IDS.
Route	The ordered list of offices a document passes through.
Hand-off	The moment a document passes from one office to another.
Endorse	To recommend a document for approval.
Return	To send a document back to the requester with reasons for correction or missing requirements.
Status	The current stage of a document (e.g., Submitted, Returned, Ready for Release).
Processing History	The list of all actions done on a document, with the office, reasons, and time.
Control Number	The unique number used to track and refer to a document.
Revision Number	The number that shows the version of a document after it is revised and resubmitted.
Request Order	The order in which requests are handled. The request submitted first is handled first, based on the date and time of submission
Priority Flag	A mark showing how urgent a request is.
Repository	The central place where all document records and files are kept.
Template	A ready-made blank form or letter for a document that requesters can download and fill in.
DA (Declaration of Action)	An MSUN form used to record the actions of the offices on a document.
University email	The official email address given by MSU at Naawan to a student or an employee. It is used to log in and to receive alerts.
SO	Special Order
INC Form	Form for incomplete grade.
TOS	Table of Specifications
IPCR/OPCR	Individual Performance Commitment and Review / Office Performance Commitment and Review.
FWL/FTL	Faculty Working Load/ Faculty Teaching Load.

FSR	Faculty Service Record
DTR	Daily Time Record
DAD	Detailed Activity Design (A college form)
TO	Travel Order
LOA	Leave for Absence (student)
PDS	Personal Data Sheet (CSC Form 212)
CSC	Civil Service Commission
HR Staff	Human Resource Staff
Manual of Operations	The CESS Manual of Operations (2025), which lists the services of the college, their requirements, steps, processing times, and persons responsible.
ARTA	Anti-Red Tape Act of 2007 (Republic Act No. 9485)
Standard Processing Time	The total time given in the Manual of Operations for a service (e.g., 4-6 working days for a Travel Order)
LET	Licensure Examination for Teachers
Web Client	The part of the system that runs in the browser.
API	Application Programming Interface. The way the Web client talks to the server.
REST/JSON	A common way to build Web APIs, and the data format used to exchange data.
HTTPS	Secure Web communication.
RBAC	Role-based access control
Row-level security (RLS)	A database feature that limits which rows a user can read or change.
Supabase	A backend-as-a-service platform built on PostgreSQL,used for the database, authentication,and file storage
Vercel	The hosting platform used to deploy the system
Agile/Kanban	The development Approach used by the team, with a Kanban board and short iterations.
ISO/IEC 25010	An International standard of software quality criteria.
TBD	To be determined.

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Appendix C: Issues List

< This is a dynamic list of the open requirements issues that remain to be resolved, including TBDs, pending decisions, information that is needed, conflicts awaiting resolution, and the like.>

No.	Issue	Needed from	Status
1	IDS route: the exact order and meaning of the IDS Principal "first and second review" (before and after Department Chairperson). The Manual of Operations does not describe the IDS route.	College Secretary, IDS Principal	Open
2	The VCAF step in the endorsement route and the budget section of the DA form are not in the approved proposal or the Manual. Decide if the VCAF is a reviewing office, an external office, or out of scope.	College Dean, College Secretary	Open
3	Full names of the FTL, FSR, and DAD forms.	College Secretary	Closed (Manual of Operations)
4	Completed list of document types and the required forms and attachments for each. Appendix B gives a first list from the Manual of Operations.	College Secretary	Partly answered
5	Control Number and Revision Number rules: automatic or entered by the College Secretary, and the exact format.	College Secretary	Open
6	Allowed file types for uploads. The proposed maximum size is 10 MB per file.	Team and client	Open
7	How accounts are made for students and faculty (created by the System Admin or self-registration with approval), and if school email is required.	College Secretary, Team	Open

8	Email service and sender address, and the number of days before a reminder is sent. The Delayed mark will use the standard processing times in the Manual.	Team,client	Open
9 ex	Homepage content and the faculty and subject list (requested in the interview, not listed in the proposal scope). Confirm the priority and if the list is public.	College Secretary	Open
10	Record retention period, and if old records (logbook, Excel, Google Drive) must be moved into the system.	College Dean, College Secretary	Open
11	Creating a record for a document received in person (walk-in)	College Secretary	Open
12	Performance targets: number of users at the same time and the expected number of records.	Tea,, client	Open
13	List of supported browsers, and software versions. (Node.js, Express.js, Tailwind CSS)	Team	Open
14	Limits of the free plans of Supabase, Vercel, and the email service (storage and email volume)	DevOps	Open
15	The routes in the interview (October 8) and in the Manual of Operations differ in some documents (e.g., FTL goes to OVCAA and OC in the interview, but only to OVCAA in the Manual; Travel Order goes to the Chancellor). Confirm which route is final.	College Dean, College Secretary	Open
16	The Manual also lists services that are not document tracking (booking and use of facilities, borrowing of equipment, consultation, LET enhancement classes). Confirm that they are out of scope for Version 1.0.	College Dean	Open
17	Student services in the Manual involve the Registrar and Cashier (e.g., INC form, release of academic documents) Confirm which student services are in Version 1.0 and how these steps are recorded.	College Secretary	Open
18	Standard processing times in the Manual are ranges and include client time. Confirm which number is used to mark a document as Delayed (upper limit is proposed).	College Secretary	Open